First-order linear recurrences

For $a_n = pa_{n-1} + q$, p≠1 gives fixed point $a^* = q/(1-p)$. Subtracting it yields $b_n = pb_{n-1}$, hence

$$a_n = p^{n-n_0}a_{n_0} + q\frac{1-p^{n-n_0}}{1-p}.$$

For p=1, $a_n = a_{n_0} + (n-n_0)q$. Specify the initial index. Integer powers use $p^0=1$, including the empty-product convention $0^0=1$, without asserting a two-variable limit.

Unrolling for $n>n_0$ gives $a_n = p^2a_{n-2} + pq + q = \cdots = p^{n-n_0}a_{n_0} + q(1 + p + \cdots + p^{n-n_0-1})$. The geometric sum gives the first formula for p≠1. For p=1 there are $n-n_0$ additions of q; p=0 gives $a_n=q$ after the initial index. Check the initial term, next term, and original recurrence. The fixed-point substitution is a convenient homogenization of the same unrolling proof.

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General forcing terms

For $a_n = pa_{n-1} + f(n)$, p≠0 allows the integrating factor

$$p^{-n}a_n - p^{-(n-1)}a_{n-1} = p^{-n}f(n).$$

Telescoping gives

$$a_n = p^{n-n_0}a_{n_0} + \sum_{k=n_0+1}^{n} p^{n-k}f(k).$$

The final formula no longer divides by p, so remains defined for p=0,1 and can be verified independently by substitution. Only the integrating-factor derivation needed p≠0. Negative p requires retaining parity signs in integer powers, rather than assuming positive-term geometric intuition.

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Variable coefficients and applications

For $a_0=1$ and $a_n=na_{n-1}$, unrolling gives $a_n=n(n-1)\dots1a_0=n!$. With $a_0=0$ every term is instead zero, showing why initial data are part of a recurrence specification.

Worst-case bubble-sort comparisons satisfy $h_1=0$,$h_n=h_{n-1}+n-1$: the first pass uses $n-1$ comparisons and leaves $n-1$ elements. Thus $h_n = \sum_{j=1}^{n-1} j = n(n-1)/2 = \Theta(n^2)$. This counts comparisons, not identical elapsed time on every machine; memory hierarchy and branch prediction still matter. Variable coefficients do not directly admit constant-coefficient characteristic roots; try products, telescoping, or substitutions and verify the actual per-round states.

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Distinct characteristic roots

For $a_n + Aa_{n-1} + Ba_{n-2} = 0$, $n\geq2$, trial r^n gives characteristic polynomial r^2+Ar+B. Distinct nonzero roots give $a_n = C_1r_1^n + C_2r_2^n$, with two initial conditions fixing constants. Check zero roots and the index domain separately.

Substitution gives $r^{n-2}(r^2 + Ar + B) = 0$, so a nonzero root supplies a solution. For nonzero distinct roots,

$$\det \begin{pmatrix} r_1^n & r_2^n \\ r_1^{n+1} & r_2^{n+1} \end{pmatrix} = r_1^n r_2^n (r_2 - r_1) \neq 0.$$

They are independent. Every solution is uniquely determined by a_0,a_1, so the solution space has dimension two. The initial equations $C_1+C_2=a_0$,$C_1r_1+C_2r_2=a_1$ have determinant $r_2-r_1\neq0$ and uniquely fit all solutions. Normalize the recurrence's signs before forming its polynomial.

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How initial data select one sequence

For $n \geq 2$, let $a_n = 5a_{n-1} - 6a_{n-2}$, with $a_0 = 1, a_1 = 4$. Roots 2,3 give $a_n = C_1 2^n + C_2 3^n$.

Initial data fix the coefficients; each column uses the same n

$a_0 = 1$, $a_1 = 4$

$C_1 + C_2 = 1$

$2C_1 + 3C_2 = 4$

$C_2 = 4 - 2 \cdot 1 = 2$

$C_1 = -1$

Two values → one solution

n	0	1	2	3
-2^n	-1	-2	-4	-8
$2 \cdot 3^n$	2	6	18	54
a_n	1	4	14	46

Add down each column: −4 + 18 = 14; −8 + 54 = 46.

Check. $a_2 = 5 \cdot 4 - 6 = 14$, $a_3 = 5 \cdot 14 - 6 \cdot 4 = 46$. No particular term is needed for this homogeneous recurrence.

Repeated, higher-order, and complex roots

A repeated nonzero root ρ gives $a_n = (C_1 + C_2 n)\rho^n$. To prove the second basis, use $A = -2\rho, B = \rho^2$:

$$n\rho^n + A(n-1)\rho^{n-1} + B(n-2)\rho^{n-2} = \rho^{n-2}[n\rho^2 - 2\rho^2(n-1) + \rho^2(n-2)] = 0.$$

The initial-value matrix $\begin{pmatrix} 1 & 0 \\ \rho & \rho \end{pmatrix}$ has determinant $\rho \neq 0$, proving independence and completeness. If $\rho = 0$, the recurrence only forces $a_n = 0$ for $n \geq 2$, leaving a_0, a_1 arbitrary. The pair $0^n, n0^n$ is not a basis.

For a nonzero root r of multiplicity m in an order-k recurrence, use $r^n, nr^n, \ldots, n^{m-1}r^n$. Handle each root's multiplicity separately; the total basis count must equal k. For zero multiplicity m_0 use prefix sequences $\delta_0, \ldots, \delta_{m_0-1}$, where $\delta_j(n) = 1$ iff $n = j$. Conjugate complex roots can be paired into real trigonometric solutions. Fibonacci's roots are irrational real numbers, not nonreal conjugates.

General proof for higher-order solutions

Write the forward recurrence as P(E)a=0, with $Ea_n=a_{n+1}$, monic degree-k P, and n≥0. The first k arbitrary values determine the rest uniquely, so dimension is k. For nonzero r and polynomial q,

$$(E - r)(r^n q(n)) = r^{n+1}\Delta q(n).$$

The difference operator lowers positive degree by one and kills constants. Thus multiplicity m makes P(E) annihilate $r^n n^j$ for j<m. For another root λ≠r,

$$(E - \lambda)(r^n q(n)) = r^n[(r - \lambda)q(n) + r\Delta q(n)].$$

On polynomials of degree below m this map is triangular with nonzero diagonal r−λ, hence invertible.

In a proposed relation among bases, apply all other roots' factors to isolate the chosen r. Other terms vanish; the surviving polynomial has undergone only invertible maps and must vanish at every nonnegative integer, hence is the zero polynomial. Repeat for each root to prove independence.

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Zero roots and dimension

The kernel of E^{m_0} is precisely sequences supported at indices 0,…,m_0−1. Each E−λ for nonzero λ is triangular on this space with nonzero diagonal −λ. The same isolation argument handles the zero-root prefix bases. All k bases are independent and hence span the full solution space. Factor P over the complex numbers; real coefficients permit conjugate pairing. Roots $\rho e^{\pm i\theta}$ give real solutions $\rho^n(A\cos n\theta + B\sin n\theta)$, with real A,B fitted from real data. For a simple nonreal conjugate pair, take $\rho > 0$ and $\sin\theta \neq 0$. If each root instead has multiplicity m, include $n^j\rho^n\cos n\theta$ and $n^j\rho^n\sin n\theta$ for every $0 \leq j < m$; two constants alone do not cover repeated complex roots.

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Nonhomogeneous recurrences and resonance

Linearity gives $a_n = a_n^{(h)} + a_n^{(p)}$. First solve the homogeneous equation, choose a particular trial matching the forcing type, substitute and compare coefficients, then add the homogeneous solution and fit initial data to the total. If its base r is a root of multiplicity m, multiply the trial by n^m.

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Which root must a trial be checked against?

Let m be the multiplicity of the root in the last column, taking m=0 when it is not a root.

Forcing	Trial before resonance	Root to check
Constant	Constant	1
Degree-d polynomial	General degree-d polynomial	1
$C r^n$, $r \neq 0$	$A r^n$	r
Degree-d polynomial times r^n, $r \neq 0$	General degree-d polynomial times r^n	r
$A\cos(n\theta) + B\sin(n\theta)$	$\alpha\cos(n\theta) + \beta\sin(n\theta)$	$e^{i\theta}$ and $e^{-i\theta}$

Multiply the entire trial by n^m at resonance. The trigonometric row assumes real coefficients and $\sin\theta \neq 0$; shifts mix sine and cosine, so retain both. For θ an integer multiple of π, the sine sequence vanishes; use the exponential row with base 1 or -1.

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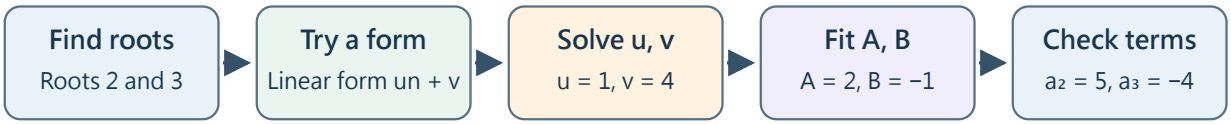
No resonance: find a particular term, then fit initial data

For $a_n=3a_{n-1}+4n, a_0=2$, the homogeneous solution is $C3^n$. Trial $An+B$ gives $An+B=(3A+4)n+(3B-3A)$, so $A=-2, B=-3$. Then $a_n=C3^n-2n-3$, and $a_0=2$ gives $C=5$. Thus $a_n=5\cdot 3^n-2n-3$, with terms 2,10,38; $3\cdot 10+8=38$.

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A second-order nonhomogeneous recurrence

Solve $a_n-5a_{n-1}+6a_{n-2}=2n+1$ for $n\geq 2$, with $a_0=5, a_1=6$. Solve the corresponding homogeneous recurrence and find one particular solution; impose the initial values on their sum.



The characteristic polynomial $r^2-5r+6=(r-2)(r-3)$ gives $a_n^{(h)}=A2^n+B3^n$. The forcing term is linear, with exponential base 1; since 1 is not a characteristic root, the trial need not be multiplied by n.

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Substitute the particular trial

Set $a_n^{(p)} = un + v$. Substitution and collection give

$$(un + v) - 5[u(n-1) + v] + 6[u(n-2) + v] = 2un + 2v - 7u.$$

Matching $2n + 1$ yields $2u = 2$ and $2v - 7u = 1$, so $u = 1, v = 4$. A particular solution is $n + 4$, and the general solution is

$$a_n = A2^n + B3^n + n + 4.$$

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Fit the initial values and verify

Apply both initial values to the total solution: $A + B + 4 = 5, 2A + 3B + 5 = 6$. Thus $A + B = 1, 2A + 3B = 1$, giving $A = 2, B = -1$:

$$\boxed{a_n = 2^{n+1} - 3^n + n + 4} \qquad (n \geq 0).$$

The recurrence independently gives

$$a_2 = 5(6) - 6(5) + 5 = 5, \qquad a_3 = 5(5) - 6(6) + 7 = -4.$$

The closed form yields $8 - 9 + 6 = 5$ and $16 - 27 + 7 = -4$ at $n = 2, 3$. For every n, the proof follows from the verified homogeneous and particular solutions and uniqueness from the two initial values.

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Resonance at a simple root

In $a_n - a_{n-1} = 2^n$, base two does not collide with root one. In $a_n - 2a_{n-1} = 2^n$ it does, requiring $Cn2^n$. For $a_n = 2a_{n-1} + 5 \cdot 2^n$, $a_0 = 3$, substitute $An2^n$:

$$An2^n = A(n-1)2^n + 5 \cdot 2^n, \qquad A = 5.$$

Thus $a_n = (C+5n)2^n$ and $C = 3$. Terms $a_1 = 16 = 2 \cdot 3 + 10$ and $a_2 = 52 = 2 \cdot 16 + 20$ check the result.

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Multiplicity two: why multiply by n^2?

The recurrence $a_n - 4a_{n-1} + 4a_{n-2} = 3 \cdot 2^n$, $n \geq 2$, has characteristic polynomial $(r-2)^2$. Divide by 2^n, putting $u_n = a_n / 2^n$.

Divide by 2^n, then test which trial polynomials vanish

$u_n - 2u_{n-1} + u_{n-2} = 3$ $(n \geq 2)$

Trial u_n	Left side after substitution	Result
A	$A - 2A + A$	0
An	$A[n - 2(n-1) + (n-2)]$	0
An^2	$A[n^2 - 2(n-1)^2 + (n-2)^2]$	$2A$

The quadratic leaves a nonzero constant: $2A = 3$, so $A = 3/2$.

Both 2^n and $n2^n$ are homogeneous solutions. A particular trial must escape their span; $An^2 2^n$ does, giving $A = 3/2$.

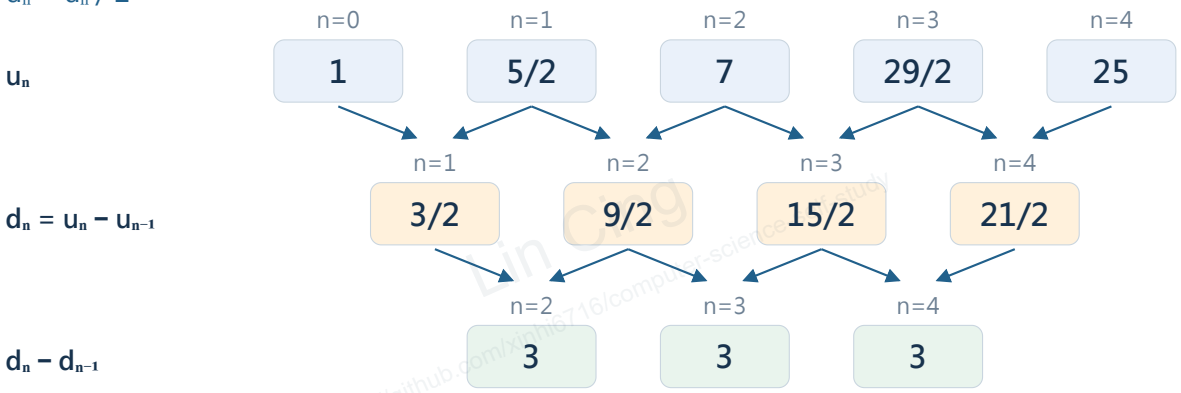
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A numerical second-difference ladder

In the same example, take $a_0 = 1, a_1 = 5$, so $u_0 = 1, u_1 = 5/2$. Compute each term from $u_n = 2u_{n-1} - u_{n-2} + 3$, then subtract adjacent entries.

Each lower entry = upper right − upper left

$u_n = a_n / 2^n$



Second differences are 3: first differences are arithmetic, and u_n is quadratic.

Reconstructing the closed form from differences

Define $d_n = u_n - u_{n-1}$ for $n \geq 1$. Then $d_1 = 5/2 - 1 = 3/2$, and $d_n - d_{n-1} = 3$ for $n \geq 2$. Hence

$$d_n = \frac{3}{2} + 3(n-1) = 3n - \frac{3}{2}, \qquad u_n = 1 + \sum_{k=1}^{n} \left(3k - \frac{3}{2}\right) = 1 + \frac{3}{2}n^2.$$

Thus $a_n = (1 + \frac{3}{2}n^2)2^n$. Equivalently, add the homogeneous term $(C_1 + C_2 n)2^n$ to the particular term $\frac{3}{2}n^2 2^n$, and fit $C_1 = 1, C_2 = 0$. **Initial data constrain the total solution, not the particular term alone.**

n	0	1	2	3	4
a_n	1	5	28	116	400

Check $a_3 = 4 \cdot 28 - 4 \cdot 5 + 3 \cdot 8 = 116$ and $a_4 = 4 \cdot 116 - 4 \cdot 28 + 3 \cdot 16 = 400$. Solution 8 also works through coefficient comparison.

Why the general resonance rule works

Let nonzero r have multiplicity s in P, and let the forcing be $r^n F(n)$, with $\deg F \le d$. The factor $(E - r)^s$ takes s differences of the polynomial part; other factors preserve degree and are invertible on fixed-degree polynomial spaces.

Restrict the trial to $r^n Q(n)$, where $Q(n) = n^s H(n)$ and $\deg H \le d$. For $0 \le j \le d$, $\Delta^s n^{s+j}$ has degree j with nonzero leading coefficient $(s + j)!/j!$. Thus the map from this trial space to degree-at-most-d polynomials is triangular with nonzero diagonal, even after the other invertible factors. It uniquely determines H for the specified forcing.

At s=0 no extra n-factor is needed. This argument assumes constant coefficients over the real or complex numbers and a nonzero base. A particular solution may still gain any homogeneous solution; uniqueness holds only within the restricted trial space. Nonlinear recurrences have no such superposition guarantee.

Generating functions and rationality

Finite-order homogeneous constant-coefficient recurrences give Q(x)A(x)=P(x), Q(0)≠0, by multiplying by x^n and summing. Thus the OGF is rational. Rational forcing OGFs preserve rationality. Linearity alone, especially with variable coefficients, does not suffice.

Conversely, if $A = P/Q$, $Q = 1 + q_1 x + \cdots + q_d x^d$, coefficient comparison past deg P gives $a_n + q_1 a_{n-1} + \cdots + q_d a_{n-d} = 0$. An eventually constant-coefficient recurrence with finitely many exceptions conversely yields a rational OGF. These are formal statements independent of convergence.

For $a_n + C_1 a_{n-1} + C_2 a_{n-2} = f_n$, n≥2, put $F(x) = \sum_{n \ge 2} f_n x^n$. The sums are $A - a_0 - a_1 x$, $x(A - a_0)$,$x^2 A$. Therefore

$$(1 + C_1 x + C_2 x^2) A = a_0 + (a_1 + C_1 a_0) x + F(x).$$

The denominator reflects recurrence coefficients; initial values and forcing determine the numerator.

The same recurrence by two methods

For $a_0=1, a_1=4, a_n=5a_{n-1}-6a_{n-2}$, roots 2,3 give $C_1+C_2=1, 2C_1+3C_2=4$, hence $C_1=-1, C_2=2$ and $a_n=-2^n+2\cdot 3^n$. The OGF route gives

$$(A-1-4x)-5x(A-1)+6x^2A=0, \qquad A=\frac{1-x}{(1-2x)(1-3x)}=-\frac{1}{1-2x}+\frac{2}{1-3x}.$$

Both methods retain the same numerator and constants.

For $a_0=1, a_n=3a_{n-1}+2$, $A-1=3xA+2x/(1-x)$, giving $A=(1+x)/((1-x)(1-3x))$. Partial fractions give $a_n=2\cdot 3^n-1$, with 1,5,17,53.

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A linear recurrence with nonrational OGF

For $h_0=0, h_n=h_{n-1}+1/n$, linearity still holds but

$$(1-x)H=\sum_{n\geq 1}x^n/n=-\log(1-x), \qquad H=-\log(1-x)/(1-x).$$

The initial terms are 0,1,3/2,11/6,25/12. Differentiating the forcing series gives $1/(1-x)$; its zero constant fixes $-\log(1-x)$.

To prove nonrationality, suppose H rational. Then $R=(1-x)H=-\log(1-x)$ is rational. At $x=1$ a rational function either has a finite limit or a pole of finite integer order $m\geq 1$. Set $t=-\log(1-x)$. As $x\to 1^-$, $t\to\infty$ but $t/(1-x)^{-m}=te^{-mt}\to 0$ for every fixed m. Logarithmic divergence is neither finite nor any pole order, contradiction. A formal rational representation would have the same Taylor expansion and analytic function, so is excluded too. Both examples require initial corrections; rationality depends on coefficients and forcing, not just linearity.

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Nonlinear recurrences

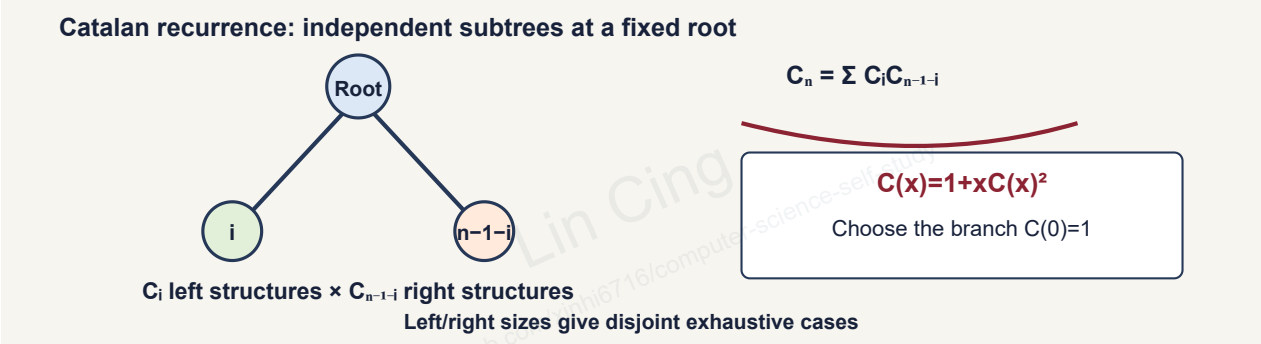
Equations such as $a_{n+1}=a_n+a_n^2$, or recurrences with products and quotients, usually have no fixed linear characteristic polynomial. Seek monotonicity, bounds, fixed points, or a reversible substitution. The change $b_n=\log a_n$ requires $a_n>0$ and converts products into sums. A nonlinear problem may ask for asymptotic bounds or invariants rather than a closed form; specify the target.

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Catalan convolution and a nonlinear closed form

Let C_n count ordered rooted binary trees on n vertices, with $C_0=1$ for the empty tree. A root uniquely joins left and right subtrees of sizes i,n−1−i, giving

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.$$



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The Catalan generating function

Summing gives C−1=xC², so xC²−C+1=0. The branch with constant one is $C = (1 - \sqrt{1-4x})/(2x)$.

Construct the formal square root directly. Let $Y(z) = \sum_{j\geq 0} \binom{1/2}{j} z^j$, with descending-product binomials. Coefficient recurrence $(j+1)c_{j+1}=(1/2-j)c_j$ gives (1+z)Y′=Y/2,Y(0)=1. For F=Y², (1+z)F′=F,F(0)=1; coefficient comparison uniquely yields F=1+z. Thus Y is the square root with constant one, not the negative branch. Substitute z=−4x and simplify descending products:

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Coefficients and the limits of the example

$$\sqrt{1-4x} = 1 - 2\sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^k.$$

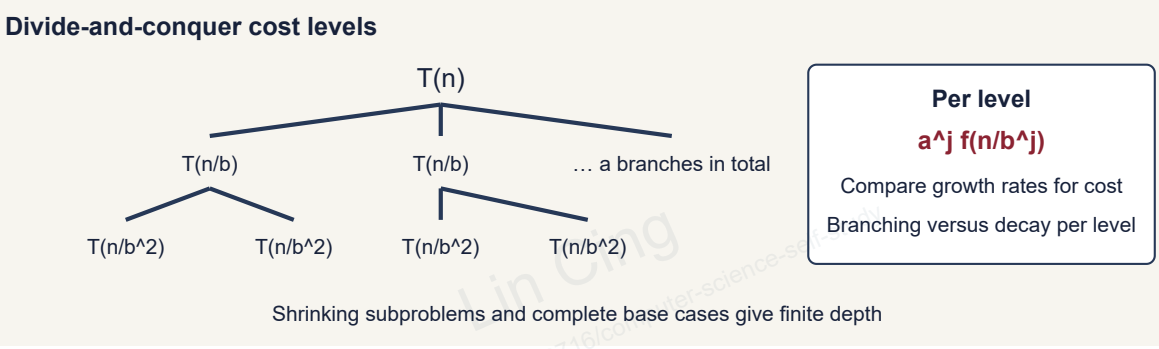
Substitution and n=k−1 give $C_n = \binom{2n}{n}/(n+1)$, with 1,1,2,5,14,42. At n=3 the five ordered trees biject with five stack outputs of {1,2,3}. Identifying left/right subtrees changes the object and invalidates the same count.

The nonlinear sequence $a_0=1, a_{n+1}=a_n+a_n^2$ gives 1,2,6,42,… and is not the Catalan recurrence. Seek transformations, convolution, or growth invariants rather than characteristic roots.

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Divide-and-conquer recurrences

For $T(n) = aT(\lceil n/b \rceil) + f(n)$, handle exact floors/ceilings and ensure subproblems strictly decrease past the base threshold. A polynomial-cost master theorem compares a with b^d for $aT(n/b)+n^d$, identifying leaves, equal-cost layers, or the root as dominant.



Expanding powers of b

Fix integers $a \geq 1, b \geq 2, c > 0$ and $n = b^k$. For $f(n) = af(n/b) + c, f(1) = c$,

$$f(b^k) = a^k f(1) + c(1 + a + \cdots + a^{k-1}).$$

If $a = 1$, this is $c(k+1) = \Theta(\log_b n)$. If $a \geq 2$ the sum is $(a^k - 1)/(a - 1)$, yielding $\Theta(a^k) = \Theta(n^{\log_b a})$. Generally,

$$f(b^k) = a^k f(1) + \sum_{j=0}^{k-1} a^j h(b^{k-j}).$$

For fixed $d \geq 0, h(n) = \Theta(n^d) > 0, f(1) = \Theta(1) > 0$, the layer ratio a/b^d gives $\Theta(n^{\log_b a})$ when $d < \log_b a$, $\Theta(n^d \log n)$ at equality, and $\Theta(n^d)$ when $d > \log_b a$. This proof is on powers of b. General integers require floor/ceiling and monotonicity bounds between adjacent powers.

A counterexample with n / log n

For T(n)=2T(n/2)+n/log₂n on n=2ᵏ≥2, a polynomial-forcing theorem cannot replace n/log n by n. Level j costs n/(k−j), so

$$T(2^k) = 2^k T(1) + 2^k \sum_{j=0}^{k-1} \frac{1}{k-j} = n(T(1) + H_k) = \Theta(n \log\log n).$$

Use a separate positive T(1); do not divide by log₂1. At n=16,T(1)=1, internal levels cost 4,16/3,8,16 and leaves cost 16, totaling 148/3=16(1+H₄). A recurrence that still calls itself at n=1 has no termination proof.

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Master-theorem examples and limits

For positive constant bases, 2T(n/2)+n gives Θ(n log n), 4T(n/2)+n gives Θ(n²), and 2T(n/4)+n gives Θ(n). Recursion-tree layer sums verify each. Nonpolynomial forcing or unsuitable a,b and division conditions require a broader theorem, recursion trees, Akra–Bazzi, or direct induction. A Θ claim requires both upper and lower bounds; O(n log n) alone does not complete a tight analysis.

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Choosing a recurrence method

Use iteration or integrating factors for first order; characteristic roots for homogeneous constant-coefficient linear equations; homogeneous plus particular solutions with resonance checks for forcing; generating functions for coefficient extraction; transformations, invariants, or bounds for nonlinear equations; and exact size recurrences followed by justified tree/theorem/induction analysis for divide and conquer. Characteristic roots outside their linear constant-coefficient family are a mathematical error, not a stylistic choice.

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Knockout tournaments

With $n=2^m$ entrants, each game eliminates one player. Every player except the champion is eliminated once, so $n-1$ games suffice and are necessary. Equivalently $T(1)=0, T(n)=2T(n/2)+1$ unrolls to $n-1$. One argument counts eliminations, the other solves a recurrence; agreement checks the base and merge-cost meanings.

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Coupled recurrences and simultaneous generating functions

For fixed matrix M and $y_n = My_{n-1} + f_n$, let $G = \sum_{n \geq 0} y_n x^n$, $F = \sum_{n \geq 1} f_n x^n$. Then

$$G - y_0 = xMG + F, \qquad G = (I - xM)^{-1}(y_0 + F).$$

No invertibility of M or analytic convergence is needed. The formal series $\Sigma x^k M^k$ multiplies $I - xM$ to I by cancellation of every positive-degree coefficient. If F has rational components, the adjugate formula and determinant constant one make G's components rational.

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Tracing the coupled state

For $a_n = 2a_{n-1} + b_{n-1}, b_n = a_{n-1} + b_{n-1}, a_0 = 1, b_0 = 0$, use the same old pair in both updates.

n	Old pair	New a_n	New b_n
1	(1,0)	2 • 1+0=2	1+0=1
2	(2,1)	2 • 2+1=5	2+1=3
3	(5,3)	2 • 5+3=13	5+3=8
4	(13,8)	2 • 13+8=34	13+8=21

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Solving the generating functions

The OGF system is

$$\begin{pmatrix} 1-2x & -x \\ -x & 1-x \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = \frac{1-x}{1-3x+x^2}, \quad B = \frac{x}{1-3x+x^2}.$$

Both components satisfy $u_n = 3u_{n-1} - u_{n-2}$ but have distinct initial values. With $r_\pm = (3 \pm \sqrt{5})/2$, partial fractions give

$$b_n = \frac{r_+^n - r_-^n}{\sqrt{5}}, \qquad a_n = \frac{(r_+ - 1)r_+^n + (1 - r_-)r_-^n}{\sqrt{5}}.$$

At n=0,1 these give (1,0),(2,1); uniqueness of each second-order recurrence then proves agreement for all terms. Elimination must preserve each component's own initial values.

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Boundaries and verification

Check initial values and substitute into the original recurrence. Verify index floors/ceilings, eventual scope of asymptotic bounds, stopping conditions, and the operation model. Exercises and solutions provide further derivations. Closed forms are one objective; lower-order terms can matter to actual cost or memory and cannot always be discarded.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 10.

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